

第五章 导数(Chapter 5 Derivatives)

本章知识点结构图(Knowledge Structure Diagram of This Chapter)

Category	Sub-item	Chinese Content	English Content
导数的概念 (Concept of Derivatives)	导数的定义 (Definition of Derivative)	导数的定义: $f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$	Definition of Derivative: $f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$
	导数的几何意义 (Geometric Meaning of Derivative)	曲线在一点处的切线的斜率	The slope of the tangent line of the curve at a point
基本初等函数的导数公式 (Derivative Formulas of Basic Elementary Functions)	常值函数导数 (Derivative of Constant Function)	常值函数导数: $C' = 0$ (C 为常数)	Derivative of Constant Function: $C' = 0$ (C is a constant)
	幂函数导数 (Derivative of Power Function)	幂函数导数: $(x^n)' = n \cdot x^{n-1}$ ($n \in \mathbf{N}$)	Derivative of Power Function: $(x^n)' = n \cdot x^{n-1}$ ($n \in \mathbf{N}$)
	正弦函数导数 (Derivative of Sine Function)	正弦函数导数: $(\sin x)' = \cos x$	Derivative of Sine Function: $(\sin x)' = \cos x$
	余弦函数导数 (Derivative of Cosine Function)	余弦函数导数: $(\cos x)' = -\sin x$	Derivative of Cosine Function: $(\cos x)' = -\sin x$
	对数函数导数 (Derivative of Logarithmic Function)	对数函数导数: $(\log_a x)' = \frac{1}{x \ln a}$ ($a > 0, a \neq 1$); $(\ln x)' = \frac{1}{x}$	Derivative of Logarithmic Function: $(\log_a x)' = \frac{1}{x \ln a}$ ($a > 0, a \neq 1$); $(\ln x)' = \frac{1}{x}$
	指数函数导数 (Derivative of Exponential Function)	指数函数导数: $(a^x)' = a^x \ln a$; $(e^x)' = e^x$	Derivative of Exponential Function: $(a^x)' = a^x \ln a$; $(e^x)' = e^x$
导数的运算法则 (Rules of Derivative)	和差导数 (Derivative of Sum and Difference)	$[u(x) + v(x)]' = u'(x) + v'(x)$ $[u(x) - v(x)]' = u'(x) - v'(x)$	$[u(x) + v(x)]' = u'(x) + v'(x)$ $[u(x) - v(x)]' = u'(x) - v'(x)$

Category	Sub-item	Chinese Content	English Content
Operation s)	乘积导数 (Derivative of Product)	$(uv)' = u'v + uv'$	$(uv)' = u'v + uv'$
	商的导数 (Derivative of Quotient)	$\left(\frac{u}{v}\right)' = \frac{uv' - uv'}{v^2} (v \neq 0)$	$\left(\frac{u}{v}\right)' = \frac{uv' - uv'}{v^2} (v \neq 0)$
	复合函数导数 (Derivative of Composite Function)	$y'_x = y'_u \cdot u'_x = f'(u) \cdot \varphi'(x)$	$y'_x = y'_u \cdot u'_x = f'(u) \cdot \varphi'(x)$
高阶导数 (Higher-Order Derivative s)	二阶导数 (Second-Order Derivative)	$f''(x)$ 是 $f(x)$ 的二阶导数, 那么 $f''(x) = (f'(x))'$	$f''(x)$ is the second-order derivative of $f(x)$, so $f''(x) = (f'(x))'$
	三阶导数 (Third-Order Derivative)	$f'''(x)$ 是 $f(x)$ 的三阶导数, 那么 $f'''(x) = (f''(x))'$	$f'''(x)$ is the third-order derivative of $f(x)$, so $f'''(x) = (f''(x))'$
	n 阶导数($n \geq 4$)(n -th Order Derivative, $n \geq 4$)	$f^{(n)}(x)$ 是 $f(x)$ 的 n 阶导数, 那么 $f^{(n)}(x) = (f^{(n-1)}(x))' (n \geq 4)$	$f^{(n)}(x)$ is the n -th order derivative of $f(x)$, so $f^{(n)}(x) = (f^{(n-1)}(x))' (n \geq 4)$
隐函数 (Implicit Function)	定义 (Definition)	以 $F(x, y) = 0$ 的形式出现的函数叫作隐函数	A function expressed in the form $F(x, y) = 0$ is called an implicit function
	求导方法 (Differentiation Method)	在方程 $F(x, y) = 0$ 两边对 x 求导, 求导过程中把 y 看作 x 的函数	Differentiate both sides of the equation $F(x, y) = 0$ with respect to x , treating y as a function of x during differentiation
参数函数 (Parametric Function)	定义 (Definition)	如果 x 和 y 之间的函数关系可以用参数方程 $\begin{cases} x = \varphi(t) \\ y = \psi(t) \end{cases}$ 确定, 那么这个函数叫作参数函数	If the functional relationship between x and y is defined by the parametric equations $\begin{cases} x = \varphi(t) \\ y = \psi(t) \end{cases}$, this function is called a parametric function
	一阶导数 (First-Order Derivative)	$\frac{dy}{dx} = \frac{\psi'(t)}{\varphi'(t)}$	$\frac{dy}{dx} = \frac{\psi'(t)}{\varphi'(t)}$
	二阶导数 (Second-Order Derivative)	$\frac{d^2y}{dx^2} = \frac{\psi''(t)\varphi'(t) - \psi'(t)\varphi''(t)}{[\varphi'(t)]^3}$	$\frac{d^2y}{dx^2} = \frac{\psi''(t)\varphi'(t) - \psi'(t)\varphi''(t)}{[\varphi'(t)]^3}$

5.1 导数的概念(Concept of Derivatives)

一、基础知识(Basic Knowledge)

(一)导数(Derivative)

- **中文定义:**函数 $y = f(x)$, 如果自变量 x 在 x_0 处有增量 Δx , 那么函数 y 相应地有增量 $\Delta y = f(x_0 + \Delta x) - f(x_0)$, 比值 $\frac{\Delta y}{\Delta x}$ 叫作函数 $y = f(x)$ 在 x_0 到 $x_0 + \Delta x$ 之间的**平均变化率**(average rate of change), 即:

$$\frac{\Delta y}{\Delta x} = \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

如果 $\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}$ 存在, 我们就说函数 $y = f(x)$ 在点 x_0 处可导 (differentiable), 这个极限叫作 $f(x)$ 在 x_0 处的导数, 记作 $f'(x_0)$ 或 $y'|_{x=x_0}$, 即:

$$f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

- **English Definition:**For the function $y = f(x)$, if the independent variable x has an increment Δx at x_0 , then the function y has a corresponding increment $\Delta y = f(x_0 + \Delta x) - f(x_0)$. The ratio $\frac{\Delta y}{\Delta x}$ is called the **average rate of change** of the function $y = f(x)$ between x_0 and $x_0 + \Delta x$, which is:

$$\frac{\Delta y}{\Delta x} = \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

If $\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}$ exists, we say the function $y = f(x)$ is **differentiable** at the point x_0 , and this limit is called the derivative of $f(x)$ at x_0 , denoted as $f'(x_0)$ or $y'|_{x=x_0}$, i.e.:

$$f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

- **求导步骤(Steps to Find Derivative):**

1) 求函数的增量(Find the increment of the function):

$$\Delta y = f(x_0 + \Delta x) - f(x_0)$$

2) 求比值(Find the ratio): $\frac{\Delta y}{\Delta x} = \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$

3) 计算极限(Calculate the limit): $\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}$, 得到导数 $f'(x_0)$

(二)导数的意义(Meaning of Derivatives)

1. 几何意义(Geometric Meaning)

- **中文:**函数 $y = f(x)$ 在点 x_0 处的导数的几何意义是:曲线 $y = f(x)$ 在点 $P(x_0, f(x_0))$ 处的切线的斜率(slope of the tangent line)。因此, 曲线 $y = f(x)$ 在点 $P(x_0, f(x_0))$ 处的切线方程(equation of the tangent line)为:

$$y - f(x_0) = f'(x_0)(x - x_0)$$

- **English:**The geometric meaning of the derivative of the function $y = f(x)$ at the point x_0 is: the **slope of the tangent line** to the curve $y = f(x)$ at the point $P(x_0, f(x_0))$. Therefore, the equation of the tangent line to the curve $y = f(x)$ at the point $P(x_0, f(x_0))$ is:

$$y - f(x_0) = f'(x_0)(x - x_0)$$

2. 物理意义(Physical Meaning)

- **中文:**一般来说, 如果物体的运动规律是 $s = s(t)$, 物体在时刻 t 的瞬时速度(instantaneous velocity) v , 就是物体在 t 到 $t + \Delta t$ 这段时间内, 当 $\Delta t \rightarrow 0$ 时平均速度的极限。

求非匀速直线运动在 t_0 时刻的瞬时速度的方法:

(1) 确定非匀速直线运动的规律(Determine the law of non-uniform linear motion): $s = s(t)$

(2) 计算时间改变量 Δt 内的位置改变量(Calculate the position change within the time change Δt): $\Delta s = s(t_0 + \Delta t) - s(t_0)$

(3) 计算瞬时速度(Calculate the instantaneous velocity):

$$v = \lim_{\Delta t \rightarrow 0} \frac{\Delta s}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{s(t_0 + \Delta t) - s(t_0)}{\Delta t}$$

- **English:** In general, if the motion law of an object is $s = s(t)$, the **instantaneous velocity** v of the object at time t is the limit of the average velocity as $\Delta t \rightarrow 0$ during the time interval from t to $t + \Delta t$.

Methods to find the instantaneous velocity of non-uniform linear motion at time t_0 :

(1) Determine the law of non-uniform linear motion: $s = s(t)$

(2) Calculate the position change within the time change Δt : $\Delta s = s(t_0 + \Delta t) - s(t_0)$

(3) Calculate the instantaneous velocity:

$$v = \lim_{\Delta t \rightarrow 0} \frac{\Delta s}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{s(t_0 + \Delta t) - s(t_0)}{\Delta t}$$

(三)基本初等函数的导数公式

(Derivative Formulas of Basic Elementary Functions)

函数类型(Function Type)	中文公式(Chinese Formula)
常值函数(Constant Function)	$C' = 0$ (C 为常数)
幂函数(Power Function)	$(x^n)' = n \cdot x^{n-1} (n \in N)$

函数类型(Function Type)	中文公式(Chinese Formula)
正弦函数(Sine Function)	$(\sin x)' = \cos x$
余弦函数(Cosine Function)	$(\cos x)' = -\sin x$
对数函数(Logarithmic Function)	$(\log_a x)' = \frac{1}{x \ln a} (a > 0, a \neq 1); (\ln x)' = \frac{1}{x}$
指数函数(Exponential Function)	$(a^x)' = a^x \ln a; (e^x)' = e^x$

5.2 导数的运算法则(Rules of Derivative Operations)

基础知识(Basic Knowledge)

运算法则(Operation Rule)	中文公式(Chinese Formula)
和差导数 (Derivative of Sum and Difference)	$[u(x) + v(x)]' = u'(x) + v'(x);$ $[u(x) - v(x)]' = u'(x) - v'(x)$
乘积导数(Derivative of Product)	$(uv)' = u'v + uv'$
商的导数(Derivative of Quotient)	$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2} (v \neq 0)$
复合函数导数 (Derivative of Composite Function)	设 $y = f(u), u = \varphi(x)$ 均可导, 则 $y'_x = y'_u \cdot u'_x = f'(u) \cdot \varphi'(x)$

5.3 高阶导数(Higher-Order Derivatives)

一、基础知识(Basic Knowledge)

(一)二阶导数(Second-Order Derivative)

- **中文定义:**如果 $\lim_{x \rightarrow x_0} \frac{f'(x) - f'(x_0)}{x - x_0}$ 存在, 我们就说函数 $f(x)$ 的导数 $f'(x)$ 在点 x_0 处可导, 这个极限叫作 $f'(x)$ 在 x_0 处的二阶导数, 记作 $f''(x_0)$ 或 $y''|_{x=x_0}$, 即:

$$f''(x_0) = \lim_{x \rightarrow x_0} \frac{f'(x) - f'(x_0)}{x - x_0}$$

- **English Definition:** If $\lim_{x \rightarrow x_0} \frac{f'(x) - f'(x_0)}{x - x_0}$ exists, we say the derivative $f'(x)$ of the function $f(x)$ is differentiable at the point x_0 , and this limit is called the second-order derivative of $f'(x)$ at x_0 , denoted as $f''(x_0)$ or $y''|_{x=x_0}$, i.e.:

$$f''(x_0) = \lim_{x \rightarrow x_0} \frac{f'(x) - f'(x_0)}{x - x_0}$$

(二)高阶导数(Higher-Order Derivatives)

- **中文定义:**二阶及二阶以上的导数都叫作高阶导数。

注:

(1) 二阶导数(Second-order derivative): $f''(x) = (f'(x))'$

(2) 三阶导数(Third-order derivative): $f'''(x) = (f''(x))'$

(3) 四阶导数(Fourth-order derivative): $f^{(4)}(x) = (f'''(x))'$

(4) n 阶导数(n -th order derivative, $n \geq 5$): $f^{(n)}(x) = (f^{(n-1)}(x))'$

函数 $y = f(x)$ 的各阶导数记作: $y', y'', y''', y^{(4)}, y^{(5)}, \dots, y^{(n)}, \dots$ 或

$f'(x), f''(x), f'''(x), f^{(4)}(x), f^{(5)}(x), \dots, f^{(n)}(x), \dots$ 或 $\frac{dy}{dx}, \frac{d^2y}{dx^2}, \frac{d^3y}{dx^3}, \dots, \frac{d^ny}{dx^n} (n = 2, 3, \dots)$ 或

$$\frac{df(x)}{dx}, \frac{d^nf(x)}{dx^n} (n = 2, 3, \dots).$$

- **English Definition:** Derivatives of the second order and above are called higher-order derivatives.

Notes:

$$(1) \text{ Second-order derivative: } f''(x) = (f'(x))'$$

$$(2) \text{ Third-order derivative: } f'''(x) = (f''(x))'$$

$$(3) \text{ Fourth-order derivative: } f^{(4)}(x) = (f'''(x))'$$

$$(4) \text{ } n\text{-th order derivative } (n \geq 5): f^{(n)}(x) = (f^{(n-1)}(x))'$$

The higher-order derivatives of the function $y = f(x)$ are denoted as:

$$y', y'', y''', y^{(4)}, y^{(5)}, \dots, y^{(n)}, \dots$$

or

$$f'(x), f''(x), f'''(x), f^{(4)}(x), f^{(5)}(x), \dots, f^{(n)}(x), \dots \text{ or } \frac{dy}{dx}, \frac{d^ny}{dx^n} (n = 2, 3, \dots) \text{ or}$$

$$\frac{df(x)}{dx}, \frac{d^nf(x)}{dx^n} (n = 2, 3, \dots).$$

5.4 隐函数的导数(Derivatives of Implicit Functions)

基础知识(Basic Knowledge)

- **显函数定义(Definition of Explicit Function):**

- 中文:以 $y = f(x)$ 的形式出现的函数叫作显函数。
- English:A function expressed in the form $y = f(x)$ is called an explicit function.

- **隐函数定义(Definition of Implicit Function):**

- 中文:以 $F(x, y) = 0$ 的形式出现的函数叫作隐函数。例如, $x - 3y^3 - 1 = 0$ 确定了一个隐函数。
- English:A function expressed in the form $F(x, y) = 0$ is called an implicit function. For example, $x - 3y^3 - 1 = 0$ defines an implicit function.

- **隐函数求导方法(Method to Find Derivative of Implicit Function):**

- 中文:在方程 $F(x, y) = 0$ 两边对 x 求导, 求导过程中把 y 看作 x 的函数。
- English:Differentiate both sides of the equation $F(x, y) = 0$ with respect to x , treating y as a function of x during the differentiation process.

5.5 参数函数的导数(Derivatives of Parametric Functions)

基础知识(Basic Knowledge)

- 参数函数定义(Definition of Parametric Function):

- 中文:如果 x 和 y 之间的函数关系可以用参数方程 $\begin{cases} x = \varphi(t) \\ y = \psi(t) \end{cases}$ 确定,

那么这个函数叫作参数函数。

- English:If the functional relationship between x and y can be

defined by the parametric equations $\begin{cases} x = \varphi(t) \\ y = \psi(t) \end{cases}$, then this function is

called a parametric function.

- 参数函数求导公式(Derivative Formulas of Parametric Functions):

1. 一阶导数(First-order derivative):

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{\psi'(t)}{\varphi'(t)}$$

2. 二阶导数(Second-order derivative):

$$\frac{d^2y}{dx^2} = \frac{\psi''(t)\varphi'(t) - \psi'(t)\varphi''(t)}{[\varphi'(t)]^3}$$